DASH: Spatio-Temporal Decision Support for High-Stress Railway Tatkal Booking

PROJECT SYNOPSIS & DOMAIN CONTEXT

Bridging the Cognitive Gap During the 10:00 AM IST IRCTC Tatkal Window

Dash transforms railway booking on the high-density **Jaipur Junction (JP) ↔ New Delhi Cluster (NDLS / DLI / DEE / DEC)** from a high-friction tabular form into a **multi-attribute visual decision support system**. Instead of forcing passengers to serially scan 20+ trains and cryptic waitlist strings under extreme time pressure, Dash externalizes working memory into a **3D/4D Pareto-Temporal Landscape**, **1-Click Natural Language Intent Parsing**, and **Counterfactual Station Quota-Jumping**.

87.5 / 100 Lab 9 Mean SUS (Grade A+)	**-54.7%** Booking Task Time Reduction
140 / 140 Automated Unit Tests Passing	**98 / 100** Lighthouse / axe Accessibility

OBJECTIVE 01 · INPUT ERGONOMICS

Zero-Friction Express Intent Entry

Eliminate 6-field form navigation before 10:00 AM using a dual-engine NLP & voice bar that configures route, date, classes, and departure windows in <5 ms.

OBJECTIVE 02 · SPATIAL & TEMPORAL INFOVIS

3D Pareto Space & 4D Checkout Forecast

Plot Fare × Duration × Confirmation Odds in 3D WebGL and simulate 60-second UPI checkout seat drain so travelers pick trains that survive payment latency.

OBJECTIVE 03 · INSTITUTIONAL ARBITRAGE

Automated Backup & Quota-Jumping

Coordinate a 1-click Plan A + Backup B pair and expose legal upstream General Quota pools (Ajmer `AII` → Boarding at Jaipur `JP`) when local `RLWL` exhausts.

TEAM MEMBER 01 · LEAD VISUALIZATION & HCI ARCHITECTURE

Harshit (`@iArceus10`)

3D/4D InfoVis & NLP

- **3D/4D WebGL Tatkal Decision Landscape (`components/TrainViz3D/`):** Engineered the Three.js coordinate cage, Pareto-optimal halo shaders, interactive weight sliders, and camera controller.
- **4D Temporal Depletion Scrubber & Plan A + B Tether (`temporalModel.ts`):** Modeled class-specific seat drain velocity (+60s UPI / +2m / +3m horizons), 3D quota-jump vector arcs, and Focus Top-3 mode.
- **Dual-Engine Express Intent Bar (`lib/intent.ts`, `ExpressIntentBar.tsx`):** Built the deterministic NLP/Hinglish parser + Gemini/Groq fallback and 140-assertion unit test suite.
- **HCI Evaluation & Accessibility Compliance:** Conducted WCAG AA contrast verification (14.2:1 Light / 12.8:1 Dark) and authored the Lab 7 & Lab 9 validation studies.

TEAM MEMBER 02 · BACKEND AI, API PIPELINE & 2D MODULES

Uttam Kumar (`@uttamshs`)

AI Copilot & Systems

- **Bounded-Context AI Copilot Backend (`app/api/copilot/`, `copilotPrompt.ts`):** Built the OpenAI/Gemini proxy with strict anti-hallucination system prompts, IP rate-limiting, and heuristic fallback.
- **Railway API Pipeline & IST Window Logic (`app/api/availability/`, `search/`):** Integrated RailKit & NTES schedule providers with 15-minute in-memory caching, timeout guards, and IST timezone calculation.
- **2D Pareto Canvas & Waitlist Probability Engine (`ParetoDecisionCanvas.tsx`, `probabilityModel.ts`):** Implemented direct-manipulation budget/duration sliders and Gigerenzer frequency framing meters.
- **Anti-Bot Verification & Group Booking Flow (`AntiBotCanvasModal.tsx`):** Built the kinematic human verification slider, progressive token queue, and senior lower-berth allocator.

Goals: Problem Statement & Research Objectives

PROBLEM STATEMENT · WHY HIGH-STRESS BOOKINGS FAIL

Cognitive Paralysis Under Radical Temporal & Probabilistic Uncertainty

At **10:00:00 AM IST** (AC) and **11:00:00 AM IST** (Sleeper), millions of passengers compete for a small pool of Tatkal berths that deplete within **90 to 120 seconds**. Existing booking interfaces (IRCTC, MakeMyTrip, ConfirmTkt) fail users not merely due to server load, but due to four fundamental **human-computer interaction bottlenecks**:

B1 **Serial Tabular Scanning & Choice Overload (Hick-Hyman Law)**
Conventional portals display 15–25 trains in flat 1D rows. Users must mentally cross-reference departure time, duration, fare, Delhi terminal (`NDLS/DEC/DEE/DLI`), and class codes in working memory—wasting 35–45 seconds before clicking.

B2 **Static Seat Counts Ignore 60-Second UPI Payment Latency**
Seeing `AVL 18` at 10:00:05 AM is misleading if that train drains at 0.45 seats/sec—by the time the passenger enters passenger details and approves a UPI mandate at +60s, the train has flipped to `TQWL`.

B3 **Opaque Waitlist Codes & Hidden Institutional Quota Partitions**
Non-expert travelers cannot calibrate the real clearance risk of `RLWL 9/WL 4` vs. `GNWL 18/WL 6`, nor can they see that booking from upstream origin **Ajmer (`AII`)** with boarding at **Jaipur (`JP`)** unlocks confirmed seats.

B4 **Hostile Anti-Bot CAPTCHAs & Single-Train Lock-In**
Illegible text CAPTCHAs penalize human vision at checkout, and when a primary train sells out mid-checkout, users must restart their search from scratch with no pre-locked backup.

RESEARCH QUESTIONS & SYSTEM OBJECTIVES

Four Measurable HCI & InfoVis Design Goals

Each system module was designed to test a specific cognitive ergonomics hypothesis and validated across iterative formative (Lab 7, $n = 7$) and summative (Lab 9, $n = 5$) evaluations:

GOAL ID	HCI / INFOVIS PRINCIPLE	TARGET SYSTEM CAPABILITY & SUCCESS METRIC
`Goal G1` **Spatial Trade-Off Encoding**	**Shneiderman's Mantra** & Pre-attentive Visual Processing (Position + Hue + Halo)	Map Fare (*X*), Duration (*Z*), and Confirmation Probability (*Y* + Color) into coordinated **3D WebGL & 2D Pareto views** to cut train selection time below **6.0 s** (achieved **3.80 s**).
`Goal G2` **4D Checkout Survival**	**Predictive Ecological Displays** & Cognitive Offloading under Time Pressure	Project seat depletion forward to **+60s (UPI checkout)**, **+2m**, and **+3m** with velocity drop-tails and auto-tethered **Plan A + Backup B** pairs on distinct trains.
`Goal G3` **Uncertainty & Quota Graph**	**Gigerenzer Frequency Framing** & Norman's Gulf of Evaluation	Translate raw `WL/RAC` codes into calibrated frequency bars ("*8 of 10 confirmed*") and expose 1-click **Ajmer (`AII`) → Jaipur (`JP`) Quota-Jumping** (+24% to +49% odds).
`Goal G4` **Express Intent & Accessibility**	**Fitts's Law**, **Miller's Law** (Focus Top 3) & **WCAG 2.1 AA**	Compress 6 search controls into 1-sentence NLP/Voice entry, achieve **SUS ≥ 80** (achieved **87.5**), **Lighthouse ≥ 95** (achieved **98**), and zero task errors.

Methodology & Technical Implementation

END-TO-END COGNITIVE DECISION WORKFLOW (4-STEP REACTIVE STATE MACHINE IN APP/PAGE.TSX)

`Stage 1 · Intent Capture`

Dual-Engine Express Intent Bar

Deterministic regex/token parser (<5ms) + optional Gemini 2.5 Flash / Groq LLM fallback extracts station, date, classes & time window with live verification chips.

`Stage 2 · Visual Exploration`

3D/4D WebGL & 2D Pareto Views

Coordinated Three.js isometric space, 4D checkout forecast scrubber (`0s/+60s/+2m/+3m`), 1-click Ajmer Quota Boost arcs, Focus Top-3 filter, and Linked 2D Table.

`Stage 3 · Group Allocation`

Multi-Passenger & Berth Engine

Detects senior citizens (60+) & children (<12) to auto-assign `Lower Berth` preferences, keep families in the same coach, and retain Backup Train B on standby.

`Stage 4 · Verification & Handoff`

Kinematic Anti-Bot & Queue

Replaces distorted text CAPTCHAs with a 1.5s human drag-docking canvas challenge, progressive concurrency queue, seat-drain velocity bar, and PNR receipt modal.

MODULE 01 · 3D/4D MATHEMATICAL MODEL

Multi-Criteria Scoring & Temporal Drain

Implemented in `scoringEngine.ts` & `temporalModel.ts`:

- **Min-Max Normalized Composite Score:**

  ```
  S_i = w_s·P_i(t) + w_f·(1 - F̂_i) + w_d·(1 - D̂_i)
  ```

  where weights (w_s, w_f, w_d) rebalance dynamically on $\Sigma w = 1$ and Pareto dominance filters non-dominated frontiers.
- **4D Seat Depletion Forecast:** Models class-specific burn rates $v_c \in [1.8, 12.5]$ seats/min with opening-surge multiplier (1.35× for $t \le 60$s) to compute **+60s UPI Checkout Survival** and render 3D velocity drop-tails.
- **Auto-Backup Pair Coordination:** Selects highest-ranked `Plan A` and an independent `Backup B` on a distinct train number with ≥ 60% survival at +60s.

MODULE 02 · ALGORITHMIC QUOTA & NLP

Quota-Jumping Graph & Trust Sanitization

Implemented in `quotaJumping.ts` & `lib/intent.ts`:

- **Upstream Origin Quota Arbitrage:** Maps the Western Railway trunk (`AII → KSG → FL → JP → AWR → RE → NDLS`). When Jaipur's Remote Location pool (`RLWL/PQWL`) saturates on #12957, #12016, #20977, or #12413, switching ticket origin to **Ajmer (`AII`)** with boarding at **Jaipur (`JP`)** taps the master `GNWL` pool (+₹65 to +₹260 delta; legal under IRCTC Circular 2018/TG-I/20/P).
- **Strict LLM Trust-Boundary Sanitizer:** Every free-form output from `/api/ai-intent` passes through `sanitizeLlmIntent()` & `normalizeBookingIntent()`—neutralizing XSS/injection payloads, clamping dates to the 14-day horizon, and preserving local deterministic precedence.

MODULE 03 · BACKEND & API RESILIENCE

3-Tier Availability & Bounded Copilot

Implemented in `app/api/{availability,copilot,search,live}`:

- **Zero-Failure 3-Tier Data Pipeline:** 1. *Primary:* Live **RailKit SDK** & **RailRadar** APIs with `AbortController` timeouts (1.8s–3.5s) and 15-min in-memory caching.
  2. *Secondary:* Public Indian Railway API with automatic cooldown circuit-breakers.
  3. *Baseline:* Verified Northern/North-Western Railway corridor schedule with explicit provenance badges (`Live` vs. `Timetable` vs. `Estimated`).
- **Bounded-Context AI Copilot (`/api/copilot`):** Supplies live `DATA` JSON to OpenAI/Gemini endpoints under strict anti-hallucination rules, IP rate-limiting, and seamless server-side heuristic fallback.

Verification: 32 Scoring/Temporal/Quota Unit Tests + 108 NLP Intent & Sanitization Tests = 140/140 Passing

Results & Evaluation (Lab 7 Baseline vs. Lab 9 & Final Build)

TASK 1 · FIND & SELECT TRAIN

3.80 s (-36.7%)

Down from **6.00 s** (SD dropped `3.61s → 0.98s`); 100% completion (5/5).

TASK 2 · GROUP & REVIEW FLOW

6.40 s (-54.7%)

Down from **14.14 s** (SD dropped `19.33s → 1.54s`); 0 errors / 0 assistance.

SYSTEM USABILITY SCALE (SUS)

87.5 / 100 (+15.0)

Up from **72.5** in Lab 7 (+20.7% relative gain); **Grade A+ (Excellent)**.

ACCESSIBILITY & UNIT TESTS

98/100 · 140/140

Lighthouse/axe 98/100 (0 critical/serious); WCAG AA **14.2:1** Light / **12.8:1** Dark.

EVALUATION MEASURE	LAB 7 BASELINE (N = 7)	LAB 9 VALIDATION (N = 5)	OBSERVED IMPACT & INTERPRETATION
Sampling & Protocol	Convenience cohort `P05–P11`	Independent cohort `P01–P05`	Zero participant overlap eliminated learning carryover effects.
Task 1 Time (Search & Pick)	Mean **6.00 s** (SD 3.61 s, range 2.0–11.0 s)	Mean **3.80 s** (SD 0.98 s, range 2.8–5.1 s)	**-36.7% mean duration**; 73% tighter SD via quick chips & Smart Match.
Task 2 Time (Passengers)	Mean **14.14 s** (SD 19.33 s, range 1.0–53.0 s)	Mean **6.40 s** (SD 1.54 s, range 4.9–8.2 s)	**-54.7% mean duration**; senior/group berth automation removed outliers.
Errors & Assistance	0 failures; **4/7 asked verbal help** on fare/WL	**0 errors; 0 help requests** across all 5 sessions	Provenance badges & trade-off pills enabled 100% autonomous completion.
SUS Usability Score	Mean **72.5** (Grade B; 60.0–87.5)	Mean **87.5** (Grade A+; 80.0–92.5)	**+15.0 pt jump** into industry top-decile usability benchmark.

PARTICIPANT ID	TASK 1 COMPLETE / TIME	TASK 2 COMPLETE / TIME	ERRORS / COACHING	INDIVIDUAL SUS SCORE
`P01 (N1)`	Yes · **2.8 s**	Yes · **4.9 s**	0 / None	`92.5 / 100`
`P02 (N2)`	Yes · **4.6 s**	Yes · **7.8 s**	0 / None	`85.0 / 100`
`P03 (N3)`	Yes · **5.1 s**	Yes · **8.2 s**	0 / None	`80.0 / 100`
`P04 (N4)` & `P05 (N5)`	Yes · **3.4 s** & **3.1 s**	Yes · **6.1 s** & **5.0 s**	0 / None	`90.0` & `90.0`

ITERATIVE ENGINEERING PROGRESSION · LAB 7 → LAB 9 → FINAL RELEASE

All 9 Usability Issues & Lab 9 "Next Steps" Closed in Final Build

- **Lab 7 Issues Resolved (8/9 in Lab 9 + 9th in Final Build):** 14-day quick date chips (`Resolved`), progressive card disclosure (`Resolved`), Delhi terminal cluster presets (`Resolved`), Smart Match rationale pills (`Resolved`), explicit Live vs. Timetable provenance badges (`Resolved`), senior lower-berth group automation (`Resolved`), simulated payment boundary & PNR receipt (`Resolved`), and default Light Mode + Dark toggle (`Resolved`).
- **Lab 9 Remaining Item Closed — Dynamic Sold-Out Re-Routing & Weight Sliders:** Lab 9 identified two final-release targets: (a) interactive user weighting sliders and (b) automated fallback re-routing when a quota class sells out. Both are now fully implemented via the **3D Weight Slider Panel** (`WeightSliderPanel.tsx`) and **Auto-Backup Pair (Plan A + Backup B) + 1-Click Filter Relaxation** (`IntentStrip.tsx`).
- **Multi-Modal Accessibility Verification:** Verified on Chrome 129 & Firefox 131 (`1920×1080`, `1440×900`) with **NVDA 2024.3** (Windows) and **VoiceOver** (macOS). Full `Tab/Shift+Tab/Escape` focus trapping, `aria-live="polite"` filter announcements, and programmatic contrast assertions (`scoringEngine.test.ts` Test 6).

All 5 Predefined Quantitative Targets Met: T1 ≥ 90% (100%), T1 < 6.0s (3.80s), T2 ≥ 90% (100%), T2 < 8.0s (6.40s), SUS ≥ 80.0 (87.5)

Findings, Insights, and Key Learnings

KEY FINDING 01 · TEMPORAL ERGONOMICS

Seat Depletion Velocity Matters More Than Static Seat Count

In high-concurrency Tatkal windows, displaying a static `AVL 20` at 10:00 AM misleads users when popular rakes drain at >10 seats/min. Projecting survival to **+60s (UPI Checkout completion)** and rendering **3D velocity drop-tails** immediately steered users away from high-burn traps toward trains that remain confirmable through payment.

KEY FINDING 02 · PHASE-DEPENDENT MODALITY

Conversational AI Excels Pre-Window; 1-Click Pills Win at 10:00 AM

Our evaluation revealed a sharp split by temporal phase: during the **09:55 AM pre-planning phase**, users valued the **Dash AI Copilot** and **Express Intent NLP bar** to explore trade-offs. However, once the **10:00 AM countdown started**, typing became cognitive overhead—users relied exclusively on **1-click** `Focus Top 3`, `+60s UPI`, and `Book Plan A + B` buttons.

KEY FINDING 03 · INSTITUTIONAL TRANSPARENCY

Visualizing Topology Bridges Norman's Gulf of Evaluation

None of our baseline participants knew that booking from **Ajmer (`AII`)** with boarding at **Jaipur (`JP`)** could legally bypass Jaipur's exhausted `RLWL` pool. Representing the corridor as a node-link topology map with 3D quota-jump arcs turned an obscure railway rule into an intuitive 1-click decision (+₹130–₹160 for +24% to +49% confirmation gain).

KEY FINDING 04 · PROVENANCE & TRUST CALIBRATION

Explicit Data Provenance Eliminates Prototype Confusion

In Lab 7, 4 of 7 participants questioned whether seat counts were live or illustrative. Introducing persistent three-state provenance badges (`Live check`, `Timetable`, `Estimated`), Gigerenzer frequency framing ("*8 of 10 confirmed*"), and an explicit simulated-payment boundary reduced participant hesitation and verbal clarification requests to **zero** in Lab 9.

METHODOLOGICAL BOUNDARIES & LIMITATIONS

Honest Scope of the Academic Prototype

- **Corridor & Sample Scope:** Evaluated on the Jaipur (`JP`) ↔ Delhi cluster with $N = 12$ total student/commuter sessions across Labs 7 & 9 on desktop viewports (`1920×1080` / `1440×900`).
- **Reservation & Payment Boundary:** Dash is a decision-support research prototype, not an authorized IRCTC PRS ticketing agent; fallback corridor inventories and PNR checkout handoffs are explicitly labeled as demonstrative when upstream live APIs are rate-limited.

REAL-WORLD UTILITY & KEY TAKEAWAY

A Plug-and-Play Decision Layer for Public Ticketing Infrastructure

Dash's modules—**Express Intent**, **3D/4D Checkout Survival**, **Auto-Backup A+B**, and **Quota-Jumping**—operate purely on standard schedule and availability payloads. They can be deployed directly as a client-side decision-support layer over **IRCTC NextGen, ConfirmTkt, or RailOne** to make high-stress public digital infrastructure calmer, fairer, and human-centered.

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Thank You · Questions & Live Demonstration